

Native Plant Seed Sowing Guide

Western Australian native species – Joondalup, WA

Version 1.1 · September 2026 · 23 species

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How to use this guide

Each species has its own page set, listed alphabetically by scientific name. Every page follows the same layout: quick facts, a sowing calendar, a plant snapshot, seed collection and storage, the sowing and growing-on method, common problems, and a blank record table so the nursery can log germination and refine the advice over time.

Sowing windows are written for the Joondalup / Perth coastal climate. Most south-west WA seed germinates best at cool to mild temperatures (roughly 15–25°C), so autumn is the main sowing season. Germination figures come from routine tests at Alcoa's Marrinup Nursery and are a guide to what a good seed lot can do – not a guarantee.

Species index – at a glance

Species	Page	Difficulty	Pre-treatment	Best sowing	Depth	Germ. %*	Seed / 100 plants*
<i>Allocasuarina fraseriana</i> Sheoak / Western Sheoak	6	Easy	None	Mar – May	2–3 mm	37%	1.3 g
<i>Allocasuarina humilis</i> Dwarf Sheoak	8	Easy-Moderate	None	Mar – May	2–3 mm	25%	1.3 g
<i>Anigozanthos manglesii</i> Red and Green Kangaroo Paw	10	Moderate	Dry heat 100°C 3 h, or aged seed	Mar – May	1–2 mm	70%	0.5 g
<i>Austrostipa flavescens</i> Coast Speargrass	12	Easy-Moderate	Smoke water; aged seed	Mar – May	0–2 mm	–	–
<i>Banksia attenuata</i> Slender Banksia / Candle Banksia	14	Easy-Moderate	None (extract from cones)	Mar – May	3–5 mm	60%	6.7 g
<i>Banksia menziesii</i> Firewood Banksia	16	Easy-Moderate	None (extract from cones)	Mar – May	3–5 mm	56%	3.7 g
<i>Banksia sessilis</i> Parrot Bush	18	Easy-Moderate	None (smoke water optional)	Mar – May	3–5 mm	42%	3.0 g
<i>Bossiaea eriocarpa</i> Common Brown Pea	20	Easy-Moderate	Boiling water 30 s	Mar – May	2–3 mm	44%	0.4 g
<i>Calothamnus quadrifidus</i> One-sided Bottlebrush	22	Easy	None	Mar – Apr, Sep – Oct	Surface	26%	0.2 g
<i>Corymbia calophylla</i> Marri	24	Easy	None	Mar – Apr, Sep – Oct	About 5 mm	60%	5.3 g
<i>Daviesia nudiflora</i>	26	Moderate	Boiling water 30 s	Mar – May	3–5 mm	22%	1.8 g
<i>Dianella revoluta</i> Blueberry Lily / Flax Lily	28	Difficult	GA 24 h + smoke water 24 h	Mar – Apr	3–5 mm	<1%†	–
<i>Eucalyptus gomphocephala</i> Tuart	30	Easy	None	Mar – Apr, Sep – Oct	1–2 mm	42%	0.4 g
<i>Eucalyptus marginata</i> Jarrah	32	Easy	None	Mar – Apr, Sep – Oct	2–3 mm	31%	3.0 g
<i>Gompholobium tomentosum</i> Hairy Yellow Pea	34	Easy-Moderate	Dry heat or hot water	Mar – May	3–4 mm	34%	0.8 g
<i>Hovea pungens</i> Devil's Pins	36	Easy-Moderate	Boiling water 30 s	Mar – May	4–6 mm	41%	3.0 g
<i>Kennedia prostrata</i> Scarlet Runner / Running Postman	38	Easy	Hot water + smoke water	Mar – May	About 5 mm	30%	8.3 g
<i>Melaleuca huegelii</i> Chenille Honeymyrtle	40	Easy	None	Mar – Apr, Sep – Oct	Surface	58%	0.1 g
<i>Melaleuca systema</i> Coastal Honeymyrtle	42	Easy-Moderate	None	Mar – Apr, Sep – Oct	Surface	14%	0.3 g

Species	Page	Difficulty	Pre-treatment	Best sowing	Depth	Germ. %*	Seed / 100 plants*
<i>Olearia axillaris</i> Coastal Daisybush	44	Easy-Moderate	Air-dry 3 months; smoke water	Aug – Sep	Surface	21%	1.4 g
<i>Petrophile linearis</i> Pixie Mops	46	Difficult	Remove hairs, nick coat; smoke	Mar – May	3–5 mm	<1%†	–
<i>Petrophile macrostachya</i>	48	Difficult *	Remove hairs, nick coat; smoke	Mar – May	3–5 mm	–	–
<i>Spyridium globulosum</i> Basket Bush	50	Moderate	Hot water (or dry heat)	Apr – Jun	2–3 mm (needs dark)	41%	0.5 g

* Alcoa Marrinup Nursery routine test averages (after the pre-treatment shown). 'Seed / 100 plants' is the approximate weight of seed that produced 100 seedlings in those tests – sow extra to allow for losses. † Untreated seed – germinates poorly without the pre-treatment shown. '–' = no nursery test data.

Sowing calendar

Species	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
<i>Allocasuarina fraseriana</i>	–	OK	Best	Best	Best	OK	OK	OK	OK	OK	–	–
<i>Allocasuarina humilis</i>	–	OK	Best	Best	Best	OK	OK	OK	OK	OK	–	–
<i>Anigozanthos manglesii</i>	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
<i>Austrostipa flavescens</i>	–	OK	Best	Best	Best	OK	OK	OK	–	–	–	–
<i>Banksia attenuata</i>	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–
<i>Banksia menziesii</i>	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–
<i>Banksia sessilis</i>	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
<i>Bossiaea eriocarpa</i>	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
<i>Calothamnus quadrifidus</i>	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–
<i>Corymbia calophylla</i>	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–
<i>Daviesia nudiflora</i>	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
<i>Dianella revoluta</i>	–	OK	Best	Best	OK	–	–	–	–	–	–	–
<i>Eucalyptus gomphocephala</i>	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–
<i>Eucalyptus marginata</i>	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–
<i>Gompholobium tomentosum</i>	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
<i>Hovea pungens</i>	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
<i>Kennedia prostrata</i>	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
<i>Melaleuca huegelii</i>	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–
<i>Melaleuca systema</i>	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–
<i>Olearia axillaris</i>	–	–	OK	OK	OK	OK	OK	Best	Best	OK	–	–
<i>Petrophile linearis</i>	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
<i>Petrophile macrostachya</i>	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
<i>Spyridium globulosum</i>	–	–	OK	Best	Best	Best	OK	OK	–	–	–	–

Perth season

Summer

Autumn

Winter

Spring

Sum.

 Best window

 Workable – slower or less reliable

 Avoid / low success

Work back from your planting date: most of these species need roughly 3–6 months from sowing to a plantable tube.

Pre-treatments explained

Many WA species have seed dormancy that must be broken before they will germinate. The species pages say which treatment to use.

Treatment	How to do it	Used for
None	Sow untreated seed. Remove from cones or follicles first where needed.	<i>Allocasuarina</i> , <i>Banksia</i>
Boiling water (30 s)	Drop seed into boiling water for 30 seconds, then drain and cool (Alcoa Marrinup method).	Hard-seeded peas, e.g. <i>Daviesia</i> , <i>Hovea</i>
Hot water soak	Pour near-boiling water over the seed (at least 10 parts water to 1 part seed) and leave to cool and soak overnight. Sow the seed that has swollen (1.5–2× its size); re-treat the rest.	Hard-seeded peas
Dry heat	Spread seed on a tray in a pre-heated oven. Peas: 80–120°C for 10 min to 1 hour. Kangaroo paw: 100°C for 3 hours.	<i>Gompholobium</i> , <i>Anigozanthos</i>
Smoke water	Soak seed for up to 24 hours in diluted smoke water (e.g. 1:9), or water trays with it after sowing. Also has slight antifungal properties.	Fire-responsive species, e.g. <i>Anigozanthos</i> (optional for others)
Aged seed	Store seed dry at room temperature for one or more years before sowing – dormancy fades with time (after-ripening).	<i>Anigozanthos</i>

Safety: boiling water and hot ovens scald – use tongs, strainers, heat-proof containers and gloves.

Difficulty ratings

Rating	Meaning
Easy	Germinates reliably with the standard method; few losses.
Easy–Moderate	Germinates well, but one factor needs care (e.g. damping off, extraction, low fill).
Moderate	Needs the right pre-treatment and timing; germination may be low or variable.
Moderate–Difficult	Low or erratic germination; specialist treatment usually needed.
Difficult	Very low germination from seed; cuttings or other methods usually preferred.

Confidence

Level	Meaning
High	Species-specific germination data and handling guidance are published.
Medium	Some species data (e.g. few tests) plus genus-level guidance. Verify with trials.
Low	Little information available; treat as a trial protocol.

Standard sowing method

Species pages refer back to these steps and only note where a species differs.

Step	What to do
1. Seed mix	Sterile, free-draining, low-nutrient mix: about 80–85% washed river sand + 15–20% coir or peat , or a commercial native seed-raising mix. Use fresh mix and clean trays. Pre-moisten before sowing.
2. Check the seed	Label every lot (species, provenance, collection date). Cut-test a few seeds – good seed is firm and white inside. For peas, a float test removes damaged seed.
3. Pre-treat	Use the treatment on the species page (see Pre-treatments explained). Sow soon after hot-water treatment.
4. Sow	Cover to about twice the seed's thickness . Fine seed: sprinkle on the surface and press in lightly, or cover with a thin layer of fine sand or vermiculite. Large seed (e.g. <i>Banksia</i> , peas): sow direct into tubes to avoid pricking-out damage.
5. Water in (post-treatment)	Water with a fine spray – smoke water is a useful first watering (slight antifungal effect, more even germination). Keep moist, never wet. A capillary bed lets you water from below.
6. Germinate	Shaded, sheltered, protected from rain and drying wind. Check daily. Watch for damping off, snails and slugs.
7. Prick out	When seedlings can be handled – usually when the second pair of leaves appears; earlier for taprooted species. Lift with a narrow spatula, hold by the leaves, not the stem, and firm mix around the roots with a gentle spray.
8. Shade house	About 2 weeks in the shade house after pricking out. Don't let tubes dry out (a capillary bed helps).
9. Harden off	Move gradually into full sun over 2–4 weeks . Pot on to larger containers after 1–3 months if needed; don't let roots become pot-bound.

Seed storage

Topic	Guidance
General rule	Dry the seed (air-dry or desiccator, never an oven), seal it in labelled airtight containers and keep it cool and dark, away from vermin. A sealed fridge is good; freezing suits many species but not all.
Exceptions	<i>Anigozanthos manglesii</i> lost germination at –18°C – store at room temperature or cooler, not frozen. <i>Allocasuarina</i> and <i>Banksia</i> seed is short-lived once out of the cone – store in the cone, or dry and cool.
Long-lived	Hard-coated pea seed keeps many years dry (related <i>Acacia</i> seed has stayed viable 50+ years).
Collecting	Collecting from bushland generally needs a DBCA licence and landholder permission. Collect only ripe seed from many plants to keep genetic diversity.

Hygiene and damping off

Damping off (*Pythium*, *Rhizoctonia*) rots seedlings at soil level, especially in wet, crowded trays. Prevent it with sterile mix, clean trays, thin sowing, good airflow and bottom watering. Remove affected seedlings quickly and use a registered fungicide only if the problem persists. Keep pots off the ground to reduce *Phytophthora* risk.

Allocasuarina fraseriana

Sheoak / Western Sheoak

Family
Casuarinaceae

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	None	Mar – May	2–3 mm	37%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	OK	OK	–	–
Best window												
Workable – slower or less reliable												
Avoid / low success												

Plant snapshot

Habit	Erect tree, 5–15 m. Male and female flowers usually on separate trees (dioecious).
Flowering	May – Oct. Woody cones up to 40 mm long follow on female trees.
Habitat	Lateritic soils and white, grey or yellow sand; jarrah forest and sand dunes.
Local note	Recorded in the Wanneroo LGA. Only female trees bear cones – note which trees you collect from.

Seed collection & storage

Item	Details
Seed collection	Cones mature within about a year of flowering. Collect cones that have turned pale to dark brown with the valves still closed . Use secateurs or a pole pruner.
Extraction & cleaning	Place cones in a paper bag or tray in a warm, dry spot – they open readily and drop the winged seeds (samaras). Sieve off cone debris.
Seed quality	Many seeds look sound but are empty. Cut-test a sample before sowing and discard lots with empty or shrivelled seed.
Storage life	Short once released from the cone unless kept dry and cool. Store dried seed sealed in the fridge, or keep seed in closed cones until the sowing season.
Typical germination	About 37% (untreated, Alcoa nursery average). About 76 seedlings per gram – roughly 1.3 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	Mar – May (cooling autumn weather, 15–25°C). Feb and Jun – Oct are workable. Avoid Nov – Jan (heat).
Seed mix	Standard sterile native seed mix (see Standard method).
Pre-treatment	None needed . Seed becomes sticky (mucilaginous) when wet – mix dry seed with a little dry, sterile yellow sand so it spreads evenly and can be seen.
Sowing depth & method	Scatter thinly and cover lightly with about 2–3 mm of fine sand or vermiculite. Direct sowing into tubes or cells (a few seeds each, thinned later) reduces losses – sheoak seedlings are prone to damping off after pricking out.
After sowing	Water in with a fine spray (smoke water is optional – aerosol smoke gave similar results to untreated seed in Alcoa trials). Keep moist, not wet. Watch for damping off.
Germination time	Seedlings emerge after about 2–5 weeks .
Pricking out	If sown in trays, prick out early – once seedlings are about 2–3 cm tall and can be handled – disturbing roots as little as possible.

Item	Details
Shade house & hardening	About 2 weeks in the shade house after pricking out or thinning, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse at soil level	Damping off. Sow thinly, use sterile mix, water from below where possible, direct sow into tubes.
Very low germination	Seed lot mostly empty. Cut-test before sowing; collect from closed, mature cones.
Seed clumps together	Mucilage. Mix dry seed with dry sand before sowing; don't pre-soak.

Notes

Other notes	Cuttings are rarely used. Seed from several trees keeps genetic diversity in revegetation stock.
Confidence	High. Published nursery test data and WA seed-handling notes are available.
Sources	See References: 1, 2, 3, 4, 5

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

Allocasuarina humilis

Dwarf Sheoak

Family
Casuarinaceae

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy-Moderate	None	Mar – May	2–3 mm	25%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	OK	OK	–	–

Best window

Workable – slower or less reliable

Avoid / low success

Plant snapshot

Habit	Erect or spreading shrub, 0.2–2 m. Dioecious or monoecious.
Flowering	May – Nov.
Habitat	Sand (often over laterite), sandy clay and gravel on sandplains and dunes.
Local note	Widespread in the South West. Collect from local-provenance plants and record the source.

Seed collection & storage

Item	Details
Seed collection	Collect mature brown cones with valves still closed , usually within a year of flowering. On dioecious plants only female plants bear cones.
Extraction & cleaning	Dry cones in a paper bag in a warm, dry place until they open; sieve out the small winged seeds.
Seed quality	Germination is often low – cut-test a sample. Many seeds can be empty.
Storage life	Short once released from the cone unless kept dry and cool. Store sealed in the fridge, or keep in closed cones until needed.
Typical germination	About 25% (untreated, Alcoa nursery average). About 77 seedlings per gram – roughly 1.3 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	Mar – May (15–25°C). Feb and Jun – Oct are workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	None needed. Mix dry seed with dry, sterile sand for even spreading (seed turns sticky when wet).
Sowing depth & method	Cover lightly, about 2–3 mm . Direct sow a few seeds per tube/cell and thin, or sow thinly in trays. Sow more generously than for <i>A. fraseriana</i> to allow for lower germination.
After sowing	Water in with a fine spray; keep moist, not wet. Good airflow to limit damping off.
Germination time	About 2–5 weeks .
Pricking out	Prick out early, at about 2–3 cm tall, with minimal root disturbance.
Shade house & hardening	About 2 weeks in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse at soil level	Damping off. Sow thinly, sterile mix, avoid overwatering.
Very low germination	Empty seed. Cut-test; collect from closed, fully mature cones.
Seed clumps together	Mix dry seed with dry sand; don't pre-soak.

Notes

Other notes	Slower-growing shrub – allow extra time to reach planting size.
Confidence	High. Published nursery test data and WA seed-handling notes are available.
Sources	See References: 1, 2, 3, 4, 5

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

Anigozanthos manglesii

Red and Green Kangaroo Paw

Family
Haemodoraceae

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Dry heat 100°C 3 h, or aged seed	Mar – May	1–2 mm	70%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Best window												
Workable – slower or less reliable												
Avoid / low success												

Plant snapshot

Habit	Rhizomatous perennial herb, 0.2–1.1 m (flower stems). Often short-lived – best treated as a 2–3 year plant.
Flowering	Aug – Nov (green and red). WA's floral emblem.
Habitat	White, yellow or grey sand and sandy loam.
Local note	Recorded in the Wanneroo LGA. Also called Mangles Kangaroo Paw.

Seed collection & storage

Item	Details
Seed collection	Seed ripens roughly three months after flowering (about Nov – Feb). Collect when capsules dry and brown, before they split and shed.
Extraction & cleaning	Dry capsules in paper bags; rub and sieve to separate the small seed.
Seed quality	Fresh seed is strongly dormant – untreated germination can be very low (about 1% in one Alcoa trial).
Storage life	Stores well for several years dry at room temperature, and germination improves with age (after-ripening). Research found freezing (–18°C) reduced germination – store dry at 23°C or cooler, not in the freezer.
Typical germination	About 70% after heat treatment (Alcoa nursery average). About 211 seedlings per gram – roughly 0.5 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	Mar – May – germinates in cool conditions (tested at 15–20°C). Jun – Aug workable (slower). Avoid Sep – Feb.
Seed mix	Standard sterile native seed mix; very free-draining.
Pre-treatment	Choose one: • Dry heat: spread seed on a tray in an oven at 100°C for 3 hours (Geraldton community nursery; Alcoa). • Aged seed: use seed stored at room temperature for a few years. • Smoke water (e.g. 24 h soak at 1:9) – response improves the longer seed has been stored.
Sowing depth & method	Sow thinly on the surface of trays and cover very lightly, about 1–2 mm of fine sand or vermiculite.
After sowing	Water in with a fine spray (smoke water optional). Keep moist but not wet. Protect from snails and slugs.
Germination time	About 3–6 weeks (around 20 days in the Geraldton nursery).

Item	Details
Pricking out	When the second leaf appears and seedlings can be handled (about 2–3 cm). Lift gently with a narrow spatula.
Shade house & hardening	About 2 weeks in the shade house after pricking out, then move gradually to full sun over 2–4 weeks. Keep foliage dry and airflow good.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Almost no germination	Seed still dormant. Heat-treat, use older seed, or add smoke water.
Black spots spreading on leaves	Ink disease (fungal). Remove affected plants, improve airflow, avoid wetting foliage.
Seedlings chewed	Snails and slugs. Bait or hand-remove.

Notes

Other notes	Division of clumps also works. Seed-grown plants vary slightly; keep local provenance for revegetation.
Confidence	High. Well researched: dormancy, heat, smoke and storage responses are published.
Sources	See References: 1, 3, 5, 6, 7, 8, 9

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

Austrostipa flavescens

Coast Speargrass

Family
Poaceae

DIFFICULTY	PRE-TREATMENT				BEST SOWING			SOWING DEPTH		TYPICAL GERMINATION			
Easy-Moderate	Smoke water; aged seed				Mar – May			0–2 mm		Not recorded			
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
Sowing	–	OK	Best	Best	Best	OK	OK	OK	–	–	–	–	
Best window Workable – slower or less reliable Avoid / low success													

Plant snapshot

Habit	Tufted perennial grass, 0.2–1.2 m.
Flowering	Sep – Oct.
Habitat	Grey, red or calcareous sand, sandy loam and limestone; sand dunes.
Local note	Coastal species. Collect from local-provenance stands and record the source.

Seed collection & storage

Item	Details
Seed collection	Seed ripens in late spring to early summer after flowering. Collect when seed heads turn straw-coloured and seed pulls away easily; it sheds quickly once ripe.
Extraction & cleaning	Strip seed by hand and dry in paper bags. Long awns can be trimmed to make sowing easier.
Seed quality	Fresh seed is often dormant (after-ripening needed).
Storage life	Store dry at room temperature. Older seed (1–2 years) gives higher germination in <i>Austrostipa</i> (Geraldton community nursery).
Typical germination	No Alcoa data for this species. A related speargrass (<i>A. caespitosa</i>) gave 73% after aerosol smoke.

Sowing & growing on

Item	Details
Best sowing time	Mar – May (autumn). Feb and Jun – Aug workable. Avoid Sep – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	- Use seed stored for a year or more, or - Water trays with smoke water after sowing (Geraldton method). - With fresh seed only, gibberellic acid plus smoke water can lift germination.
Sowing depth & method	Sow very shallowly – on or just below the surface (0–2 mm), pressed in lightly. Seed needs light to germinate.
After sowing	Water in with smoke water; keep moist.
Germination time	About 21 days (Geraldton nursery).
Pricking out	Prick out or divide small clumps when seedlings are 3–5 cm with several leaves.
Shade house & hardening	About 2 weeks in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Poor germination from fresh seed	Dormant. Store seed a year, or use smoke water (and GA if available).
Seed buried too deep	Needs light. Sow on the surface and press in.
Seedlings rot	Overwatering. Keep moist, not wet.

Notes

Other notes	Can also be multiplied by dividing established clumps.
Confidence	Medium. WA community-nursery method and genus-level dormancy research; no Alcoa data for this species.
Sources	See References: 3, 5, 23

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

Banksia attenuata

Slender Banksia / Candle Banksia

Family
Proteaceae

DIFFICULTY	PRE-TREATMENT				BEST SOWING		SOWING DEPTH		TYPICAL GERMINATION			
Easy-Moderate	None (extract from cones)				Mar – May		3–5 mm		60%			
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–
Best window Workable – slower or less reliable Avoid / low success												

Plant snapshot

Habit	Lignotuberous tree or shrub, usually 2–4 m (up to 10 m). Resprouts after fire.
Flowering	Oct – Feb (yellow candles).
Habitat	White, yellow, brown or pale red sand, sometimes over laterite; sand dunes and sandplains.
Local note	Recorded in Joondalup and Wanneroo. Highly susceptible to <i>Phytophthora</i> dieback – strict hygiene.

Seed collection & storage

Item	Details
Seed collection	Follicles can take a year or more to mature. Collect old, dry, dark grey-brown cones with closed follicles. Cones hold seed for some time, so collection isn't tied to one month.
Extraction & cleaning	Open follicles with heat: oven at 120–130°C for 1 hour (or 200°C for 10 min), or briefly heat then soak in water for several hours and dry in a warm oven. Separate the winged black seed from the woody separator. Good seed is white and firm inside. Never microwave cones.
Seed quality	Seed from older cones tends to have lower viability than from younger cones – favour recent cones.
Storage life	Short-lived once extracted unless stored dry and cool. Best kept in the cone until the sowing season; extracted seed can remain viable for some years sealed, dry and cool (10–25°C, 40–60% RH).
Typical germination	About 60% (untreated, Alcoa nursery average). About 15 seedlings per gram – roughly 7 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	Mar – May – sow once the weather cools after March. Jun – Sep workable. Avoid Oct – Feb (heat raises damping-off losses).
Seed mix	Free-draining, pathogen-free, low-phosphorus mix (pasteurise home-made mix at 60°C for 30 min). Slightly acid (about pH 6).
Pre-treatment	None needed once seed is out of the follicle.
Sowing depth & method	Direct sow 1–2 seeds per deep tube (preferred), or sow thinly in deep trays. Just cover with fine mix or vermiculite (about 3–5 mm).
After sowing	Water in with a fine spray. This species is highly vulnerable to damping off – water from below (capillary bed) if possible, or use a registered fungicide at germination (e.g. furalaxyl or captan, as per label).
Germination time	About 16–49 days .

Item	Details
Pricking out	If tray-sown, prick out at the cotyledon stage, no later than a few weeks after germination , before the taproot lengthens.
Shade house & hardening	About 2 weeks under shade cloth after pricking out, then harden off gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse soon after emerging	Damping off. Sterile mix, bottom watering, fungicide at germination, less crowding.
Yellowing, dying seedlings in pots	Phosphorus toxicity or <i>Phytophthora</i> . Use very low-P fertiliser; keep pots off the ground.
Few seeds from cones	Immature or old cones. Collect ripe, recent cones; check follicles have developed.

Notes

Other notes	Feed only with half-strength, very low-phosphorus fertiliser once established (Proteaceae are P-sensitive).
Confidence	High. Published nursery test data and Banksia seed-handling guidance are available.
Sources	See References: 1, 3, 10, 11, 12, 13

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

Banksia menziesii

Firewood Banksia

Family
Proteaceae

DIFFICULTY	PRE-TREATMENT				BEST SOWING		SOWING DEPTH		TYPICAL GERMINATION			
Easy-Moderate	None (extract from cones)				Mar – May		3–5 mm		56%			
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	OK	–	–	–
Best window Workable – slower or less reliable Avoid / low success												

Plant snapshot

Habit	Tree or shrub, 1.3–7 m, from a lignotuber. Resprouts after fire.
Flowering	Feb – Oct, peaking May – Jul (pink, red or yellow).
Habitat	White, grey or yellow sand.
Local note	Recorded in Joondalup and Wanneroo. Medium to high susceptibility to <i>Phytophthora</i> dieback.

Seed collection & storage

Item	Details
Seed collection	Up to 25 follicles per cone, each with 1–2 seeds. Follicles open mainly after fire (or release seed after about two years), so collect mature, closed cones .
Extraction & cleaning	Heat cones to open follicles: oven at 120–130°C for 1 hour (or 200°C for 10 min), or briefly heat, soak for several hours, then dry in a warm oven. Separate seed from the woody separators.
Seed quality	Good seed is white and firm with a dark coat. Recent cones generally give better viability than old cones.
Storage life	Short-lived once extracted unless stored dry and cool. Keep in the cone until needed where possible.
Typical germination	About 56% (untreated, Alcoa nursery average). About 27 seedlings per gram – roughly 3.7 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	Mar – May , once the weather cools. Jun – Sep workable. Avoid Oct – Feb.
Seed mix	Free-draining, pathogen-free, low-phosphorus mix (about pH 6).
Pre-treatment	None needed once seed is out of the follicle.
Sowing depth & method	Direct sow 1–2 seeds per deep tube (preferred), or thinly in deep trays. Just cover (about 3–5 mm).
After sowing	Water in with a fine spray; bottom watering where possible. Watch for damping off; fungicide at germination if needed (as per label).
Germination time	About 26–40 days .
Pricking out	If tray-sown, prick out at the cotyledon stage within a few weeks of germination.
Shade house & hardening	About 2 weeks under shade cloth after pricking out, then harden off gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse soon after emerging	Damping off. Sterile mix, bottom watering, less crowding.
Yellowing or dying in pots	Phosphorus toxicity or <i>Phytophthora</i> . Very low-P feed; strict hygiene; pots off the ground.
Follicles won't open	Not enough heat. Repeat heating, or heat then soak and re-dry.

Notes

Other notes	Takes about five to seven years to flower from seed.
Confidence	High. Published nursery test data and Banksia seed-handling guidance are available.
Sources	See References: 1, 3, 10, 11, 12, 13

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

Banksia sessilis

Parrot Bush

Family
Proteaceae

DIFFICULTY	PRE-TREATMENT			BEST SOWING			SOWING DEPTH			TYPICAL GERMINATION		
Easy-Moderate	None (smoke water optional)			Mar – May			3–5 mm			42%		
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Best window Workable – slower or less reliable Avoid / low success												

Plant snapshot

Habit	Upright, prickly shrub or small tree to 6 m. Killed by fire and regrows from seed.
Flowering	Mainly Jul – Nov (cream-yellow domes).
Habitat	Sandy soils over laterite or limestone.
Local note	The Swan Coastal Plain form is var. <i>cygnorum</i> . Formerly <i>Dryandra sessilis</i> .

Seed collection & storage

Item	Details
Seed collection	Not serotinous: most seed is released in autumn even without fire. Collect ripe heads in summer, before release.
Extraction & cleaning	Put heads in a paper bag in a warm place or a low oven (no hotter than 60°C) until the follicles open; cut stubborn follicles with secateurs.
Seed quality	Good seed is firm and white inside.
Storage life	Seed in the soil is short-lived, but cleaned seed keeps well stored dry and cool .
Typical germination	About 42% (untreated, Alcoa nursery average). About 33 seedlings per gram – roughly 3 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	Mar – May – sow once the weather cools after March. Jun – Aug workable. Avoid Sep – Feb.
Seed mix	Free-draining, sterile, low-phosphorus mix.
Pre-treatment	None needed. Geraldton soaks seed in smoke water (1:9) for 24 hours.
Sowing depth & method	Direct sow 1–2 seeds per tube, just covered (about 3–5 mm).
After sowing	Water in; avoid excess water – damping off can be a problem.
Germination time	About 2–6 weeks (14 days at Geraldton).
Pricking out	If tray-sown, prick out at the cotyledon stage, within a few weeks of germination.
Shade house & hardening	About 2 weeks in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse	Damping off. Water less, sterile mix, good airflow.
Few seeds in heads	Collected too late – seed already released in autumn. Collect in summer.
Yellowing seedlings	Phosphorus toxicity. Use very low-P fertiliser.

Notes

Other notes	Cuttings have proved virtually impossible – seed is the only practical method. Flowers within about two years.
Confidence	High. Nursery test data (as <i>Dryandra sessilis</i>) and WA seed-handling notes available.
Sources	See References: 3, 5, 20, 25

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

Bossiaea eriocarpa

Common Brown Pea

Family
Fabaceae

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy-Moderate	Boiling water 30 s	Mar – May	2–3 mm	44%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
	Best window	Workable – slower or less reliable								Avoid / low success		

Plant snapshot

Habit	Erect or straggly, spreading shrub, 0.2–1 m.
Flowering	Jul – Nov (yellow and brown-red pea flowers).
Habitat	Usually sandy soils.
Local note	Recorded in Joondalup and Wanneroo.

Seed collection & storage

Item	Details
Seed collection	Pods ripen in summer and shed seed when ripe. Bag developing pods with stockings or muslin.
Extraction & cleaning	Rub and sieve the small seed from pod fragments.
Seed quality	Hard-seeded – fresh seed is dormant until the coat is treated. Float test to remove damaged seed.
Storage life	Long. Keeps many years when dry (typical of hard-coated pea seed).
Typical germination	About 44% after boiling-water treatment (Alcoa nursery average). About 229 seedlings per gram – under 0.5 g per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	Mar – May and Sep. Feb, Jun – Aug and Oct workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	Boiling water for 30 seconds (Alcoa method), or pour near-boiling water over and soak overnight. Sow swollen seed; re-treat the rest.
Sowing depth & method	Small seed – cover about 2–3 mm . Direct sow a few seeds per tube and thin.
After sowing	Water in; keep moist, not wet.
Germination time	Can start about one week after treatment.
Pricking out	At the cotyledon to first true leaf stage, before the taproot lengthens.
Shade house & hardening	About 2 weeks in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Seed coat not broken. Re-treat unswollen seed.
Swollen seed rots	Over-boiled or too wet. Keep to 30 s; water less.
Seedlings collapse	Damping off. Sterile mix, airflow, avoid crowding.

Notes

Other notes	Very small seed for a pea – a little goes a long way.
Confidence	High. Nursery germination data (40 tests) and WA hard-seed guidance available.
Sources	See References: 3, 5, 14, 15, 16

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

Calothamnus quadrifidus

One-sided Bottlebrush

Family
Myrtaceae

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	None	Mar – Apr, Sep – Oct	Surface	26%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–

Best window	Workable – slower or less reliable	Avoid / low success
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Plant snapshot

Habit	Erect, compact or spreading shrub, usually 0.9–2 m.
Flowering	Jun – Dec (red, one-sided brushes).
Habitat	Wide variety of soils and habitats.
Local note	Recorded in the Wanneroo LGA. Several subspecies exist – label stock by subspecies and provenance.

Seed collection & storage

Item	Details
Seed collection	Woody capsules stay on the older stems for years. Collect capsules from older wood (at least a year old) at any time of year.
Extraction & cleaning	Place capsules in an open tray or paper bag in a warm spot; fine seed is released in a few days. No need to separate seed from chaff.
Seed quality	Very fine seed mixed with chaff – a cut-test and count gives seeds per gram.
Storage life	Long if kept dry and cool (typical of capsule-fruited Myrtaceae).
Typical germination	About 26% (untreated, Alcoa nursery average). About 572 seedlings per gram – a pinch goes a long way.

Sowing & growing on

Item	Details
Best sowing time	Mar – Apr and Sep – Oct (15–25°C). Feb, May – Aug and Nov workable. Avoid Dec – Jan.
Seed mix	Fine-textured, sterile native seed mix with a smooth surface.
Pre-treatment	None needed.
Sowing depth & method	Mix seed with dry, sterile yellow sand and sprinkle from a spice jar. Sow on the surface and press in lightly; don't bury.
After sowing	Water in with smoke water (Geraldton method) using a very fine spray, or water from below (bog method) so seed isn't washed away.
Germination time	About 7–10 days (Geraldton nursery).
Pricking out	When the second pair of leaves appears (about 2 cm). Seedlings are tiny – handle by the leaves.
Shade house & hardening	About 2 weeks in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings too crowded	Sown too thickly. Dilute seed well with sand.
Seed washed into clumps	Overhead watering too coarse. Use a fine mist or bottom watering.
Damping off	Crowding and wet mix. Thin sowing, airflow, sterile mix.

Notes

Other notes	Also grows readily from cuttings if a particular form is wanted.
Confidence	High. Large nursery data set (39 tests) and WA community-nursery method available.
Sources	See References: 3, 4, 5, 18

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

Corymbia calophylla

Marri

Family
Myrtaceae

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	None	Mar – Apr, Sep – Oct	About 5 mm	60%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–

Best window	Workable – slower or less reliable	Avoid / low success
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Plant snapshot

Habit	Tree to 40 m (rarely a mallee), with rough, tessellated bark.
Flowering	Dec or Jan – May (white or pink).
Habitat	Wide range: clay loam, sandy clay, gravel, grey sand over limestone, granite and laterite.
Local note	Recorded in the Wanneroo LGA. Large urn-shaped gumnuts ('honky nuts').

Seed collection & storage

Item	Details
Seed collection	Fruit takes about a year to mature. Collect brown, woody gumnuts when the valves look ready to open.
Extraction & cleaning	Keep gumnuts warm and dry in a tray or paper bag until the large winged seed is released.
Seed quality	Large seed – easy to check by cutting a few open.
Storage life	Long if stored cool and dry (eucalypt seed keeps its viability a long time).
Typical germination	About 60% (untreated, Alcoa nursery average). About 19 seedlings per gram – roughly 5 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	Mar – Apr and Sep – Oct . Feb, May – Aug and Nov workable. Avoid Dec – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	None needed. A dip in dilute bleach (10%) to surface-sterilise the seed can help.
Sowing depth & method	Large seed – cover about 5 mm . Direct sow 1–2 seeds per tube (preferred) or space out in trays.
After sowing	Water in; keep moist.
Germination time	Usually within about 1–2 weeks .
Pricking out	When seedlings have two pairs of true leaves.
Shade house & hardening	About 2 weeks in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse	Damping off. Sterile mix, avoid crowding.
Leggy seedlings	Too much shade. Move to brighter light once up.
Roots circling	Kept too long in tubes. Plant out or pot on sooner.

Notes

Other notes	Fast-growing – time sowings so trees aren't held in tubes too long.
Confidence	High. Large nursery data set (66 tests) and WA eucalypt seed notes available.
Sources	See References: 3, 4, 5, 19

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

Daviesia nudiflora

No widely used common name (a bitter-pea)

Family
Fabaceae

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Moderate	Boiling water 30 s	Mar – May	3–5 mm	22%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
	Best window	Workable – slower or less reliable								Avoid / low success		

Plant snapshot

Habit	Shrub, 0.3–1.5 m.
Flowering	Jun – Sep (orange-yellow and red-brown pea flowers).
Habitat	Sandy and clayey soils and laterite; low rises and flats.
Local note	Recorded in the Wanneroo LGA.

Seed collection & storage

Item	Details
Seed collection	Pods ripen in summer and release seed as soon as they are ripe . Tie nylon stockings or muslin bags over developing pods and collect once they split.
Extraction & cleaning	Rub pods and sieve or winnow to remove debris.
Seed quality	Hard-seeded: nearly all fresh seed is dormant until the seed coat is treated. Float test: discard seed that floats.
Storage life	Long . Hard-coated pea-family seed keeps for many years when dry (related <i>Acacia</i> seed has stayed viable 50+ years).
Typical germination	About 22% after boiling-water treatment (Alcoa nursery average, few tests). About 57 seedlings per gram – roughly 1.7 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	Mar – May and Sep . Feb, Jun – Aug and Oct are workable. Avoid Nov – Jan (heat stress, damping off).
Seed mix	Standard sterile native seed mix.
Pre-treatment	Boiling water: drop seed into boiling water for 30 seconds (Alcoa method), then cool; or pour near-boiling water over seed (at least 10 parts water to 1 part seed) and leave to soak overnight. Sow seed that has swollen ; re-treat seed that hasn't.
Sowing depth & method	Cover to about twice the seed's thickness, roughly 3–5 mm . Direct sow 2–3 seeds per tube and thin, or sow in trays.
After sowing	Water in; keep moist, not wet. Sow soon after treatment – don't let swollen seed dry out.
Germination time	Can start about one week after treatment.
Pricking out	At the cotyledon to first true leaf stage – early, before the taproot lengthens.
Shade house & hardening	About 2 weeks in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.

Item	Details
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Very low germination	Seed coat not broken. Re-treat unswollen seed; try a dry-heat pulse (80–120°C) on the next batch.
Swollen seed rots	Mix too wet or seed damaged by overlong boiling. Keep boiling to 30 s; water less.
Seedlings collapse	Damping off. Sterile mix, good airflow, avoid crowding.

Notes

Other notes	Treat with care: boiling water scalds – use tongs, a strainer and heat-proof containers.
Confidence	Medium. Nursery germination data available (small number of tests); method follows WA hard-seed guidance.
Sources	See References: 1, 3, 5, 14, 15, 16

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

Blueberry Lily / Flax Lily

Family
Asphodelaceae

DIFFICULTY Difficult	PRE-TREATMENT GA 24 h + smoke water 24 h			BEST SOWING Mar – Apr		SOWING DEPTH 3–5 mm		TYPICAL GERMINATION <1% untreated				
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	OK	–	–	–	–	–	–	–
Best window												
	Workable – slower or less reliable							Avoid / low success				

Plant snapshot

Habit	Rhizomatous, tufted perennial herb, 0.3–1.5 m.
Flowering	Sep – Dec (blue-purple), followed by blue berries.
Habitat	White or yellow sand, sandy clay, laterite, loam and clay.
Local note	Recorded in Joondalup and Wanneroo (var. <i>divaricata</i> is common on the coast).

Seed collection & storage

Item	Details
Seed collection	Collect ripe, deep-blue berries in summer after flowering.
Extraction & cleaning	Soak berries in water for a few days, then rub and wash off the pulp in a sieve. Dry the shiny black seed before storing.
Seed quality	Seed is dormant (morphophysiological dormancy): untreated seed gave under 1% in Alcoa tests.
Storage life	Not well documented. Store cleaned, dried seed sealed and cool.
Typical germination	Untreated: under 1% (Alcoa, 19 tests). Treated seed does much better – record your own results.

Sowing & growing on

Item	Details
Best sowing time	Mar – Apr – seed experiences warm conditions, then germinates as the weather cools (Geraldton: not until late April). Feb and May workable.
Seed mix	Standard sterile native seed mix.
Pre-treatment	Gibberellic acid (1 part per thousand) for 24 hours, then smoke water (1:9) for 24 hours (Geraldton method). Research shows warm stratification followed by smoke also breaks dormancy.
Sowing depth & method	Cover about 3–5 mm . Sow in trays or direct into tubes.
After sowing	Keep moist and be patient – don't discard trays too early.
Germination time	About 58–75 days (Geraldton nursery).
Pricking out	When seedlings have 2–3 leaves and are large enough to handle.
Shade house & hardening	About 2 weeks in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 5–7 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Nothing after 6 weeks	Normal – germination takes 8–11 weeks. Keep trays moist.
No germination at all	Seed still dormant. Use GA + smoke water, or keep trays for a second season.
Pulp left on seed goes mouldy	Clean seed thoroughly before drying and storing.

Notes

Other notes	Division of established clumps is a quicker way to produce plants.
Confidence	Medium. WA community-nursery method and dormancy research available; nursery results vary widely.
Sources	See References: 3, 4, 5, 23

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

Eucalyptus gomphocephala

Tuart

Family
Myrtaceae

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	None	Mar – Apr, Sep – Oct	1–2 mm	42%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–
	Best window	Workable – slower or less reliable								Avoid / low success		

Plant snapshot

Habit	Tree, 10–40 m.
Flowering	Jan – Apr (white).
Habitat	Sand over limestone on the coastal plain.
Local note	Recorded in Joondalup and Wanneroo.

Seed collection & storage

Item	Details
Seed collection	Fruit takes about a year to mature. Collect brown capsules when the valves are just starting to open; many trees hold seed for a year.
Extraction & cleaning	Keep capsules warm and dry from a day to several weeks until seed drops. Seed is mixed with chaff.
Seed quality	A cut-test and count on seed and chaff estimates seeds per gram.
Storage life	Long if stored cool and dry.
Typical germination	About 42% (untreated, Alcoa nursery average). About 228 seedlings per gram – under 0.5 g per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	Mar – Apr and Sep – Oct. Feb, May – Aug and Nov workable. Avoid Dec – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	None needed. Surface-sterilising in dilute bleach (10%) can help.
Sowing depth & method	Mix with dry sand and sprinkle thinly; cover very lightly (about 1–2 mm).
After sowing	Water in with a fine spray (smoke water optional); keep moist.
Germination time	About 1–2 weeks.
Pricking out	When seedlings have two pairs of true leaves.
Shade house & hardening	About 2 weeks in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse	Damping off. Thin sowing, sterile mix, airflow.
Uneven germination	Chaff clumps. Mix seed well with sand.
Roots circling	Held too long in tubes. Time sowings to planting.

Notes

Other notes	Local coastal-plain provenance matters for revegetation – record seed source.
Confidence	High. Nursery data (16 tests) and WA eucalypt seed notes available.
Sources	See References: 3, 4, 5, 19

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

Eucalyptus marginata

Jarrah

Family
Myrtaceae

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	None	Mar – Apr, Sep – Oct	2–3 mm	31%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–

Best window

Workable – slower or less reliable

Avoid / low success

Plant snapshot

Habit	Tree to 40 m.
Flowering	Jun – Dec or Jan (white-cream to pink).
Habitat	Grey sand, clay or sandy loam and laterite; hills and rises.
Local note	Recorded in Joondalup and Wanneroo.

Seed collection & storage

Item	Details
Seed collection	Fruit takes about a year to mature. Collect brown capsules when the valves look ready to open.
Extraction & cleaning	Keep capsules warm and dry until seed is released; sieve out large debris.
Seed quality	Cut-test a sample; seed is mixed with chaff.
Storage life	Long if stored cool and dry.
Typical germination	About 31% (untreated, Alcoa nursery average). About 33 seedlings per gram – roughly 3 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	Mar – Apr and Sep – Oct . Feb, May – Aug and Nov workable. Avoid Dec – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	None needed . Surface-sterilising in dilute bleach (10%) can help.
Sowing depth & method	Cover lightly, about 2–3 mm . Direct sow a few seeds per tube and thin, or sow thinly in trays.
After sowing	Water in; keep moist.
Germination time	About 1–2 weeks .
Pricking out	When seedlings have two pairs of true leaves.
Shade house & hardening	About 2 weeks in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings collapse	Damping off. Sterile mix, avoid crowding.
Low germination	Old or immature seed. Collect mature capsules; cut-test.
Roots circling	Held too long in tubes.

Notes

Other notes	Jarraah seedlings develop a lignotuber early – a good sign of healthy stock.
Confidence	High. Large nursery data set (67 tests) and WA eucalypt seed notes available.
Sources	See References: 3, 4, 5, 19

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

Gompholobium tomentosum

Hairy Yellow Pea

Family
Fabaceae

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy-Moderate	Dry heat or hot water	Mar – May	3–4 mm	34%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–

Best window
Workable – slower or less reliable
Avoid / low success

Plant snapshot

Habit	Erect, often spindly shrub, 0.15–1 m.
Flowering	Jul – Jan (yellow pea flowers).
Habitat	White, grey or yellow sand, sandy clay and lateritic sand; coastal limestone, dunes and plains.
Local note	Recorded in Joondalup and Wanneroo.

Seed collection & storage

Item	Details
Seed collection	Pods ripen in summer and shed seed when ripe. Bag developing pods with stockings or muslin and collect once split.
Extraction & cleaning	Rub and sieve out the small seed.
Seed quality	Hard-seeded – almost all fresh seed is dormant until the coat is treated.
Storage life	Long. Keeps for many years when dry (typical of hard-coated pea-family seed).
Typical germination	About 34% after heat treatment (Alcoa nursery average). About 125 seedlings per gram – roughly 0.8 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	Mar – May and Sep. Feb, Jun – Aug and Oct workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	Either: • Dry heat shock: oven at 80–120°C for 10 min to 1 hour (Alcoa used a dry-heat pulse for this species); or • Hot water: pour near-boiling water over the seed and let it cool in the water (easiest method). Sow seed that has swollen; re-treat the rest.
Sowing depth & method	Small seed – cover about 3–4 mm . Direct sow a few seeds per tube and thin, or sow thinly in trays.
After sowing	Water in; keep moist, not wet. Sow promptly after hot-water treatment.
Germination time	Can start about one week after treatment.
Pricking out	At the cotyledon to first true leaf stage, before the taproot lengthens.
Shade house & hardening	About 2 weeks in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Seed coat not broken. Re-treat unswollen seed or switch between heat and hot-water methods.
Seed rots after sowing	Too wet. Water less and improve drainage.
Seedlings collapse	Damping off. Sterile mix, airflow, avoid crowding.

Notes

Other notes	Spindly habit – tip-pruning young plants can give a fuller tube.
Confidence	High. Nursery germination data and WA hard-seed guidance available for this species.
Sources	See References: 1, 3, 5, 14, 15, 16

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

Hovea pungens

Devil's Pins

Family
Fabaceae

DIFFICULTY	PRE-TREATMENT				BEST SOWING			SOWING DEPTH		TYPICAL GERMINATION		
Easy-Moderate	Boiling water 30 s				Mar – May			4–6 mm		41%		
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Best window												
Workable – slower or less reliable												
Avoid / low success												

Plant snapshot

Habit	Erect, prickly, broom-like shrub, 0.2–1.8 m.
Flowering	May – Nov (blue-purple).
Habitat	Wide range: shallow soils on rock, sand, laterite, gravel, clay and loam; granite outcrops, coastal limestone and sandplains.
Local note	Recorded in Joondalup and Wanneroo. Sharp leaves – wear gloves when collecting.

Seed collection & storage

Item	Details
Seed collection	Pods ripen in late spring to summer and eject seed when ripe . Bag developing pods with stockings or muslin.
Extraction & cleaning	Collect seed from the bags; sieve out pod fragments.
Seed quality	Hard-seeded – fresh seed is dormant until the coat is treated. Float test to remove damaged seed.
Storage life	Long . Keeps for many years when dry (typical of hard-coated pea-family seed).
Typical germination	About 41% after boiling-water treatment (Alcoa nursery average). About 33 seedlings per gram – roughly 3 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	Mar – May and Sep . Feb, Jun – Aug and Oct workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	Boiling water : drop into boiling water for 30 seconds (Alcoa method), then cool; or pour near-boiling water over the seed and soak overnight. Sow swollen seed; re-treat any that haven't swollen.
Sowing depth & method	Larger seed – cover about 4–6 mm (roughly twice its thickness). Direct sow 2–3 seeds per tube and thin.
After sowing	Water in; keep moist, not wet.
Germination time	Can start about one week after treatment.
Pricking out	If tray-sown, prick out at the cotyledon stage – early, as the taproot grows fast.
Shade house & hardening	About 2 weeks in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–5 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Seed coat not broken. Re-treat unswollen seed.
Swollen seed rots	Over-boiled or mix too wet. Keep to 30 s; water less.
Taproot damage at pricking out	Pricked out too late. Direct sow into tubes instead.

Notes

Other notes	Wear gloves – the pungent (sharp) leaves make handling seedlings and stock plants prickly.
Confidence	High. Nursery germination data available for this species; method follows WA hard-seed guidance.
Sources	See References: 1, 3, 5, 14, 15, 16

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

Kennedia prostrata

Scarlet Runner / Running Postman

Family
Fabaceae

DIFFICULTY	PRE-TREATMENT				BEST SOWING		SOWING DEPTH			TYPICAL GERMINATION		
Easy	Hot water + smoke water				Mar – May		About 5 mm			30%		
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	Best	OK	OK	OK	Best	OK	–	–
Best window Workable – slower or less reliable Avoid / low success												

Plant snapshot

Habit	Prostrate or twining groundcover, spreading to about 2.5 m.
Flowering	Apr – Nov (red pea flowers).
Habitat	Usually sandy, gravelly soils.
Local note	Recorded in Joondalup and Wanneroo.

Seed collection & storage

Item	Details
Seed collection	Pods (about 50 mm) ripen in late spring to summer and split when ripe – bag them or collect as they turn brown.
Extraction & cleaning	Dry pods in paper bags and collect the seed.
Seed quality	Impervious seed coat – dormant until treated. Float test to remove damaged seed.
Storage life	Long. Seed maintains viability for extended periods.
Typical germination	About 30% after dry heat (100°C, 3 h; Alcoa). About 12 seedlings per gram – roughly 8 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	Mar – May and Sep. Feb, Jun – Aug and Oct workable. Avoid Nov – Jan.
Seed mix	Standard sterile native seed mix.
Pre-treatment	Hot water: simmer water at 95°C for 2 minutes, pour over the seed (at least 10:1 water to seed) and soak 24 hours, then soak 24 hours in smoke water (Geraldton method). Alternatives: boiling water, abrasion, or dry heat 100°C for 3 hours.
Sowing depth & method	Larger seed – cover about 5 mm . Direct sow 1–2 seeds per tube.
After sowing	Water in; keep moist, not wet.
Germination time	About 8 days (Geraldton nursery).
Pricking out	If tray-sown, prick out at the cotyledon stage.
Shade house & hardening	About 2 weeks in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Seed coat not broken. Re-treat unswollen seed.
Runners tangle in the nursery	Space tubes out; trim runners before dispatch.
Seedlings collapse	Damping off. Sterile mix, airflow.

Notes

Other notes	Cuttings of firm current season's growth also strike well.
Confidence	High. Nursery data, WA community-nursery method and ANPSA notes agree.
Sources	See References: 3, 4, 5, 14, 24

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

Melaleuca huegelii

Chenille Honeymyrtle

Family
Myrtaceae

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy	None	Mar – Apr, Sep – Oct	Surface	58%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–
	Best window	Workable – slower or less reliable								Avoid / low success		

Plant snapshot

Habit	Shrub or small tree, 0.5–5 m.
Flowering	Sep – Dec or Jan (pink, white or pink-purple).
Habitat	Sand; limestone cliffs, coastal plains and dunes.
Local note	Recorded in Joondalup and Wanneroo.

Seed collection & storage

Item	Details
Seed collection	Seed stays in the woody capsules indefinitely, so capsules can be collected at any time. Use capsules at least 12 months old.
Extraction & cleaning	Place capsules in an open container somewhere warm; fine seed is released in 2–3 days.
Seed quality	Extremely fine seed with chaff; no need to separate.
Storage life	Long if kept dry and cool; seed also keeps on the plant in closed capsules.
Typical germination	About 58% (untreated, Alcoa nursery average). About 775 seedlings per gram – a pinch is plenty.

Sowing & growing on

Item	Details
Best sowing time	Mar – Apr and Sep – Oct. Feb, May – Aug and Nov workable. Avoid Dec – Jan.
Seed mix	Fine-textured, sterile native seed mix.
Pre-treatment	None needed.
Sowing depth & method	Mix with dry, sterile yellow sand and sprinkle from a spice jar. Sow on the surface; don't cover.
After sowing	Water in with smoke water using a fine mist (Geraldton), or stand pots in a saucer of water until germination (bog method).
Germination time	About 9 days at Geraldton; 14–30 days is typical for the genus.
Pricking out	When the second pair of leaves appears. Handle tiny seedlings by the leaves.
Shade house & hardening	About 2 weeks in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Seedlings too crowded	Dilute seed more with sand.
Seed washed away	Use bottom watering or a very fine mist.
Damping off	Crowding and wet mix. Thin sowing, airflow.

Notes

Other notes	Cuttings are also reliable if a particular form is wanted.
Confidence	High. Nursery data, WA community-nursery method and ANPSA genus notes agree.
Sources	See References: 3, 4, 5, 18

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

Melaleuca systema

Coastal Honeymyrtle

Family
Myrtaceae

DIFFICULTY	PRE-TREATMENT	BEST SOWING	SOWING DEPTH	TYPICAL GERMINATION
Easy-Moderate	None	Mar – Apr, Sep – Oct	Surface	14%

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	OK	Best	Best	OK	OK	OK	OK	Best	Best	OK	–

 Best window	 Workable – slower or less reliable	 Avoid / low success
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Plant snapshot

Habit	Erect to spreading shrub, usually 0.5–2 m.
Flowering	Feb – Mar or Aug – Dec (yellow-cream).
Habitat	Sand over laterite or limestone; stabilised coastal dunes and rocky limestone.
Local note	Recorded in Joondalup and Wanneroo.

Seed collection & storage

Item	Details
Seed collection	Collect capsules at least 12 months old from older stems at any time; seed is retained in closed capsules.
Extraction & cleaning	Place capsules somewhere warm in an open container; seed is released in 2–3 days.
Seed quality	Fine seed with chaff. Germination in nursery tests was low – sow generously.
Storage life	Long if kept dry and cool.
Typical germination	About 14% (untreated, Alcoa average of only 3 tests). About 301 seedlings per gram .

Sowing & growing on

Item	Details
Best sowing time	Mar – Apr and Sep – Oct . Feb, May – Aug and Nov workable. Avoid Dec – Jan.
Seed mix	Fine-textured, sterile native seed mix.
Pre-treatment	None needed.
Sowing depth & method	Mix with dry sand and sprinkle from a spice jar. Sow on the surface ; don't cover.
After sowing	Water in with smoke water using a fine mist, or use the bog method.
Germination time	About 7 days at Geraldton; allow up to 4 weeks.
Pricking out	When the second pair of leaves appears.
Shade house & hardening	About 2 weeks in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Low germination	Empty or old seed. Use older capsules (12+ months) from vigorous plants; sow more.
Seed washed away	Use bottom watering or a very fine mist.
Damping off	Crowding and wet mix.

Notes

Other notes	Cuttings are a reliable alternative for this genus.
Confidence	Medium. Few nursery tests (3); WA community-nursery method and ANPSA genus notes available.
Sources	See References: 3, 4, 5, 18

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

Olearia axillaris

Coastal Daisybush

Family
Asteraceae

DIFFICULTY	PRE-TREATMENT				BEST SOWING			SOWING DEPTH		TYPICAL GERMINATION		
Easy-Moderate	Air-dry 3 months; smoke water				Aug – Sep			Surface		21%		
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	OK	OK	OK	OK	OK	Best	Best	OK	–	–
Best window Workable – slower or less reliable Avoid / low success												

Plant snapshot

Habit	Erect, much-branched grey shrub, 0.2–3 m.
Flowering	Jan – Jul or Nov – Dec (small white to yellow daisies).
Habitat	White, grey or red sand and loam; coastal limestone, sand dunes and rocky hillsides.
Local note	Recorded in Joondalup and Wanneroo.

Seed collection & storage

Item	Details
Seed collection	Seed usually ripens around April – May , but plants can set seed about 6 weeks after a big rain , so watch for opportunistic crops (e.g. after summer storms).
Extraction & cleaning	Collect the fluffy seed heads into paper bags.
Seed quality	Germination in nursery tests was moderate (about 21%).
Storage life	Short – needs fresh seed each year. Seed must be aired for about 3 months before storing or sowing.
Typical germination	About 21% (untreated, Alcoa nursery average). About 73 seedlings per gram – roughly 1.4 g of seed per 100 seedlings.

Sowing & growing on

Item	Details
Best sowing time	Aug – Sep – about 3 months after an autumn collection. Mar – Jul and Oct workable depending on when seed was collected. Avoid Nov – Feb.
Seed mix	Standard sterile native seed mix.
Pre-treatment	Air the seed for about 3 months after collection. Water trays with smoke water after sowing (Geraldton method).
Sowing depth & method	Sow on the surface and press in lightly – daisy-family seed often needs light to germinate.
After sowing	Water in with smoke water; keep moist.
Germination time	About 7–16 days (Geraldton nursery).
Pricking out	When the second pair of leaves appears.
Shade house & hardening	About 2 weeks in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 3–4 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
Poor germination	Old seed or not aired. Use this season's seed, aired for 3 months.
Seed buried too deep	Needs light. Surface sow.
Leggy seedlings	Too much shade. Move to full sun sooner.

Notes

Other notes	Also strikes readily from cuttings.
Confidence	High. Nursery data and a detailed WA community-nursery method available.
Sources	See References: 3, 4, 5, 23

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

Petrophile linearis

Pixie Mops

Family
Proteaceae

DIFFICULTY		PRE-TREATMENT				BEST SOWING		SOWING DEPTH		TYPICAL GERMINATION			
Difficult		Remove hairs, nick coat; smoke				Mar – May		3–5 mm		<1% untreated			
		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing		–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Best window		Workable – slower or less reliable				Avoid / low success							

Plant snapshot

Habit	Erect shrub, 0.2–1 m.
Flowering	Aug – Dec (pink and white ‘mops’).
Habitat	White, grey or yellow sand, often over laterite; sandplains.
Local note	Recorded in Joondalup and Wanneroo.

Seed collection & storage

Item	Details
Seed collection	Cones are usually ripe about six months after flowering , when they turn grey to grey-brown and lose their old floral parts.
Extraction & cleaning	Seed does not come out easily in <i>Petrophile</i> – you may need to pull the hairy nuts out of the cone with forceps.
Seed quality	Very low germination untreated (0.17% in Alcoa tests). Many nuts may be empty – cut-test.
Storage life	Not well documented. Store dry and cool.
Typical germination	Untreated: under 1% (Alcoa, 6 tests). Treated seed should do much better – record your results.

Sowing & growing on

Item	Details
Best sowing time	Mar – May (cool autumn). Jun – Aug workable. Avoid Sep – Feb.
Seed mix	Free-draining, sterile, low-phosphorus mix.
Pre-treatment	Remove the hairs and nick a small section of the seed coat to reveal the white seed (without damaging it). Smoke water (24 h soak) may also help.
Sowing depth & method	Cover about 3–5 mm . Direct sow into tubes to avoid root damage.
After sowing	Keep moist, not wet; strict hygiene.
Germination time	Can take several months – keep trays going.
Pricking out	If tray-sown, prick out at the cotyledon stage.
Shade house & hardening	About 2 weeks in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 6–8 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
No germination	Seed coat not nicked, or empty seed. Nick carefully; cut-test.
Seed damaged by nicking	Cut too deep. Remove just enough coat to see white seed.
Seedlings die in pots	Phosphorus toxicity or root rot. Low-P mix and fertiliser.

Notes

Other notes	A slow, fiddly species from seed – set larger batches and keep good records.
Confidence	Medium. Nursery tests show very low untreated germination; method follows WA Petrophile seed notes.
Sources	See References: 3, 4, 21

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

Petrophile macrostachya

No widely used common name

Family
Proteaceae

DIFFICULTY		PRE-TREATMENT				BEST SOWING		SOWING DEPTH		TYPICAL GERMINATION			
Difficult *		Remove hairs, nick coat; smoke				Mar – May		3–5 mm		Not recorded			
		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing		–	–	Best	Best	Best	OK	OK	OK	–	–	–	–
Best window		Workable – slower or less reliable							Avoid / low success				

Plant snapshot

Habit	Erect, compact or spreading, prickly shrub, 0.3–1.2 m.
Flowering	Jul – Nov (yellow-cream).
Habitat	White or yellow-brown sand and gravel; sandplains.
Local note	Recorded in the Wanneroo LGA. Prickly – wear gloves.

Seed collection & storage

Item	Details
Seed collection	Cones are usually ripe about six months after flowering, when grey to grey-brown and the old floral parts have gone.
Extraction & cleaning	Pull the hairy nuts from the cones with forceps.
Seed quality	Cut-test a sample – expect many empty nuts.
Storage life	Not well documented. Store dry and cool.
Typical germination	Not recorded. Expect low germination unless the seed coat is treated.

Sowing & growing on

Item	Details
Best sowing time	Mar – May. Jun – Aug workable. Avoid Sep – Feb.
Seed mix	Free-draining, sterile, low-phosphorus mix.
Pre-treatment	Remove the hairs and nick a small section of the seed coat; smoke water may also help.
Sowing depth & method	Cover about 3–5 mm. Direct sow into tubes.
After sowing	Keep moist, not wet; strict hygiene.
Germination time	Can take several months.
Pricking out	If tray-sown, prick out at the cotyledon stage.
Shade house & hardening	About 2 weeks in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 6–8 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
No germination	Seed coat not nicked, or empty seed.
Seed damaged by nicking	Cut less deeply.
Seedlings die in pots	Phosphorus toxicity or root rot.

Notes

Other notes	Treat as a trial – record results so this page can be updated.
Confidence	Low. No species data found. Method based on WA Petrophile seed notes and <i>P. linearis</i> results – trial protocol.
Sources	See References: 4, 21

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

Spyridium globulosum

Basket Bush

Family
Rhamnaceae

DIFFICULTY	PRE-TREATMENT			BEST SOWING			SOWING DEPTH			TYPICAL GERMINATION		
Moderate	Hot water (or dry heat)			Apr – Jun			2–3 mm (needs dark)			41%		
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sowing	–	–	OK	Best	Best	Best	OK	OK	–	–	–	–
Best window												
Workable – slower or less reliable												
Avoid / low success												

Plant snapshot

Habit	Erect shrub, usually 0.6–5 m.
Flowering	Jun – Nov (white, in woolly heads).
Habitat	Sand on coastal dunes and limestone.
Local note	Recorded in Joondalup and Wanneroo.

Seed collection & storage

Item	Details
Seed collection	Collect fruit as it ripens in late spring–summer; bag branches to catch shed seed.
Extraction & cleaning	Rub and sieve the small seed from the woolly flower heads.
Seed quality	Water-impervious seed coat (physical dormancy) – seed won't take up water until treated.
Storage life	Not well documented; hard-coated seed is usually long-lived when dry.
Typical germination	About 41% after heat treatment (Alcoa, only 2 tests). About 205 seedlings per gram . Research recorded high germination after hot water at cool (7/18°C) temperatures.

Sowing & growing on

Item	Details
Best sowing time	Apr – Jun – germinates best at winter-like temperatures (7/18°C); warmth suppresses it. Mar, Jul – Aug workable. Avoid Sep – Feb.
Seed mix	Standard sterile native seed mix.
Pre-treatment	Hot water (about 90–95°C) breaks the seed-coat dormancy – keep it short, as over-long treatment can kill seed. Alcoa used a dry-heat pulse instead.
Sowing depth & method	Cover about 2–3 mm – germination is suppressed by light, so keep the seed covered and trays shaded.
After sowing	Water in; keep moist in a cool, shaded spot.
Germination time	Not well documented – allow several weeks and record results.
Pricking out	When the second pair of leaves appears.
Shade house & hardening	About 2 weeks in the shade house after pricking out, then move gradually to full sun over 2–4 weeks.
Time to planting size	Est. 4–6 months from sowing (record your own).

Common problems

Problem	Likely cause and fix
No germination	Seed coat not broken, or trays too warm or too light. Re-treat; cover seed; sow in cooler months.
Seed killed by treatment	Hot water too long. Shorten the soak.
Damping off	Wet, crowded trays. Thin sowing, airflow.

Notes

Other notes	One of the few coastal shrubs here that needs cool temperatures <i>and</i> darkness to germinate.
Confidence	High. Species-specific dormancy research and nursery data available.
Sources	See References: 3, 4, 22, 23

Nursery records

Log each sowing here to confirm or update the timing, treatment and germination above.

Date sown	Seed lot / provenance	Treatment	Qty sown	No. up	Germ. %	Notes

References

Numbers match the source list on each species page. Web addresses shortened; accessed September 2026.

#	Source	Address
1	Australian Native Plants Society (ANPSA) – Plant Propagation: seed mix, pre-treatments, pricking out, growing on	anpsa.org.au/plant-propagation/
2	Cochrane A. (2007) Seed Notes 7 – Allocasuarina. Wildflower Society of WA / DEC Threatened Flora Seed Centre	library.dbca.wa.gov.au/FullTextFiles/PAM01994.pdf
3	Cromer E.L. (2007) Seed Germination and Research Records from Alcoa's Marrinup Nursery. Alcoa World Alumina Australia	library.dbca.wa.gov.au/static/FullTextFiles/924754.pdf
4	FloraBase, WA Herbarium (DBCA) – species profiles 1411, 1728, 1732, 1800, 1834, 2299, 2301, 3710, 3824, 3957, 3966, 4044, 4828, 5426, 5659, 5708, 5920, 8127, 11636, 17104, 17240, 18598	florabase.dbca.wa.gov.au
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7	Seed Science Research – Seed ageing of four Western Australian species in relation to storage environment (incl. <i>A. manglesii</i>)	cambridge.org (Seed Science Research)
8	AoB PLANTS (2018) – Seed dormancy regulates germination response to smoke and temperature (<i>Anigozanthos</i>)	academic.oup.com/aobpla/article/10/4/ply042/5048955
9	Wikipedia – <i>Anigozanthos manglesii</i> (cultivation, ink disease)	en.wikipedia.org/wiki/Anigozanthos_manglesii
10	ANPSA – The Propagation of <i>Banksia</i> (extraction, mix, damping off, low-P feeding, hardening)	anpsa.org.au/the-propagation-of-banksia/
11	Seed Notes 8 – <i>Banksia</i> . Wildflower Society of WA / DEC (2007)	library.dbca.wa.gov.au/FullTextFiles/PAM01995.pdf
12	Wikipedia – <i>Banksia attenuata</i> and <i>Banksia menziesii</i> (germination times, damping off, serotiny)	en.wikipedia.org
13	Crawford A.D. – Seed storage and longevity of Australian plant species (PhD thesis; <i>Banksia</i> cone age and viability)	library.dbca.wa.gov.au/static/FullTextFiles/924221.pdf
14	Seed Notes 15 – Hard seed: <i>Chorizema</i> , <i>Daviesia</i> , <i>Gastrolobium</i> and <i>Gompholobium</i> . Wildflower Society of WA / DEC	library.dbca.wa.gov.au/FullTextFiles/PAM02002.pdf
15	Seed Notes 9 – <i>Acacia</i> (hard-seed longevity, float test). Wildflower Society of WA / DEC	library.dbca.wa.gov.au/FullTextFiles/PAM01996.pdf
16	Auld T.D. (1996) Ecology of the Fabaceae: fire, ants and the soil seedbank. <i>Cunninghamia</i> 4(4)	botanicgardens.org.au
17	Seed Notes 2 – Seed collection and storage. Wildflower Society of WA / DEC (2007)	library.dbca.wa.gov.au/FullTextFiles/PAM01989.pdf
18	ANPSA – <i>Melaleuca</i> genus page (seed from old capsules, no pre-treatment, bog method, 14–30 days)	anpsa.org.au/genera/melaleuca/
19	Seed Notes 6 – <i>Eucalyptus</i> . Wildflower Society of WA / DEC (2007)	library.dbca.wa.gov.au/FullTextFiles/PAM01993.pdf
20	Seed Notes 13 – <i>Dryandra</i> (now <i>Banksia</i>). Wildflower Society of WA / DEC (2007)	library.dbca.wa.gov.au/FullTextFiles/PAM02000.pdf
21	Seed Notes 18 – <i>Isopogon</i> and <i>Petrophile</i> . Wildflower Society of WA / DEC (2007)	library.dbca.wa.gov.au/FullTextFiles/PAM02005.pdf
22	Seed Science Research – Physical dormancy in seeds of six genera of Australian Rhamnaceae (incl. <i>Spyridium globulosum</i>)	cambridge.org (Seed Science Research)
23	Florabank Guidelines Module 11 – Seed germination and dormancy. Australian Network for Plant Conservation	anpc.asn.au/wp-content/uploads/2025/08/11.-Seed-Germination-and-Dormancy.pdf
24	ANPSA – <i>Kennedia prostrata</i> profile	anpsa.org.au/plant_profiles/kennedia-prostrata/

#	Source	Address
25	Wikipedia – Banksia sessilis (seed release, fire response, cuttings)	en.wikipedia.org/wiki/Banksia_sessilis

Revision log

Version	Date	Changes
1.0	Sep 2026	First edition: Allocasuarina fraseriana, A. humilis, Anigozanthos manglesii, Banksia attenuata, B. menziesii, Daviesia nudiflora, Gompholobium tomentosum, Hovea pungens.
1.1	Sep 2026	Added 15 species: Austrostipa flavescens, Banksia sessilis, Bossiaea eriocarpa, Calothamnus quadrifidus, Corymbia calophylla, Dianella revoluta, Eucalyptus gomphocephala, E. marginata, Kennedia prostrata, Melaleuca huegelii, M. systema, Olearia axillaris, Petrophile linearis, P. macrostachya, Spyridium globulosum.